

PROJECT TITLE : SCRIPT-CONTROLLED ACL-RESTRICT RECORD ACCESS BASED ON FIELD VALUE ASSIGNED

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SCRIPT-CONTROLLED ACL – RESTRICT RECORD ACCESS BASED ON FIELD VALUE ASSIGNED

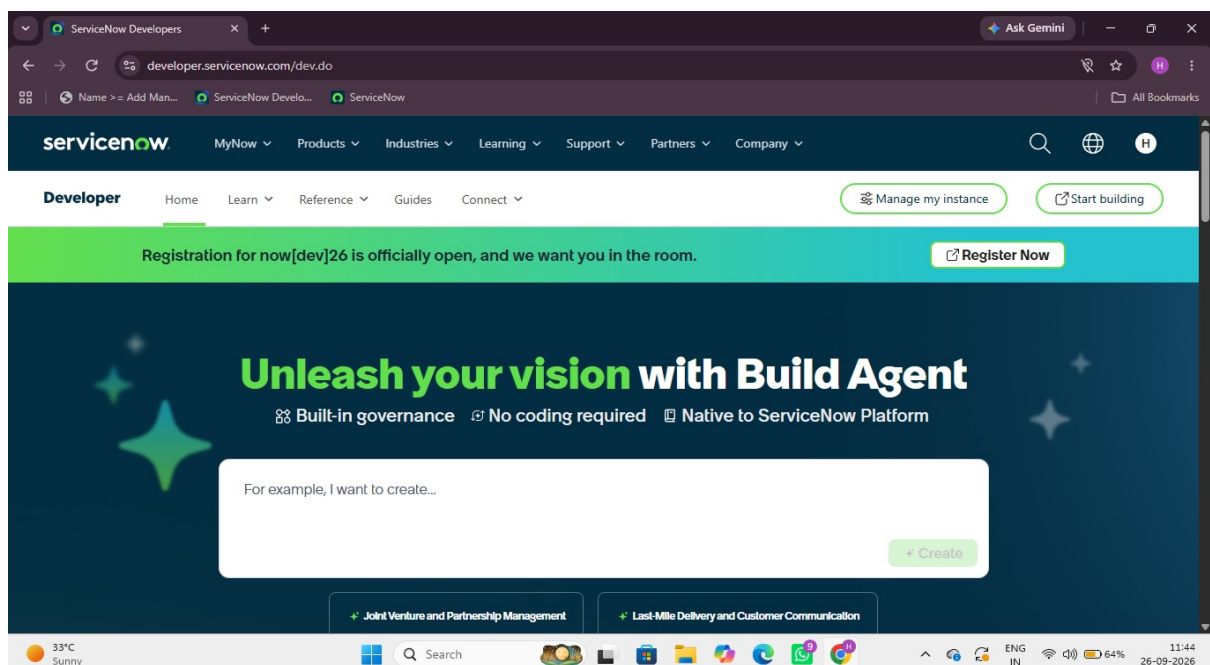
1. Introduction

Access Control List (ACL) is an important security feature in ServiceNow used to control access to records and operations. ACLs determine whether a user is allowed to read, create, write, or delete a record.

This project demonstrates a script-controlled ACL that restricts record access based on the value of a field. In this project, an Institution Details table is created with a Branch field containing values such as ECE, EEE, and CSE.

A custom role named bb1 is assigned to the EEE user. The Read ACL uses a script to control access. Users with the required role can access the records according to the configured security conditions, while administrators have complete access.

The project is implemented using ServiceNow's User Administration, Tables, and Access Control features. The PDF specifies that the project covers READ, CREATE, WRITE, and DELETE ACLs.



2. Objectives

- The main objectives of this project are:
- To understand Access Control Lists in ServiceNow.
- To create users and custom roles.
- To create a custom table for storing institution/student details.
- To control record access using ACLs.
- To restrict records based on field values.
- To implement Read, Create, Write, and Delete ACLs.
- To understand how roles and ACL scripts work together.
- To prevent unauthorized users from viewing or modifying protected records.
- To provide administrators with full access to the records.

3. Existing System

In a basic system without properly configured ACLs, users may have access to records that they are not supposed to view or modify.

For example, if student records belonging to different branches are stored in the same table, unauthorized users may be able to view or modify records belonging to another branch.

This can lead to:

- Unauthorized record viewing.

- Unauthorized modification.
- Accidental deletion.
- Lack of proper data security.
- Difficulty in controlling access based on user roles.

Therefore, a proper record-level security mechanism is required.

4. Proposed System

The proposed system uses ServiceNow ACLs with role-based and script-controlled access.

A custom table named Institution Details is created. It contains student information along with a Branch field having values such as ECE, EEE, and CSE.

Different custom roles are created:

bb1 – Read access

bb2 – Create access

bb3 – Write access

bb4 – Delete access

The PDF specifies that these roles are assigned progressively so that the required operations can be controlled separately.

The system therefore provides controlled access to records instead of giving unrestricted access.

5. System Requirements

5.1 Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Internet connection
- Keyboard and Mouse

5.2 Software Requirements

- ServiceNow Instance
- Web Browser
- Internet Connection
- Administrator Access
- ServiceNow Security Admin role for ACL configuration

The PDF specifically requires elevation of the Security_admin role before creating the ACL.

6. System Modules

The project consists of the following modules:

Module 1 – User Creation

A test user named EEE User is created through User Administration → Users.

Module 2 – Role Creation

Custom roles bb1, bb2, bb3, and bb4 are created and assigned to the required user.

Module 3 – Table Creation

A custom table called Institution Details is created with the table name:

u_institution_details

Module 4 – Field Creation

The table contains:

Student Roll Number

Student Name

Faculty Name

Branch

Email

Phone Number

Description

The Branch field is configured as a Choice field with ECE, EEE, and CSE values.

Module 5 – Record Creation

Multiple records are created with different branch values such as ECE, EEE, and CSE.

Module 6 – Read ACL

A Read ACL is created for u_institution_details. The ACL uses the bb1 role and includes a script for access control.

Module 7 – Create ACL

A Create ACL is created using the bb2 role.

Module 8 – Write ACL

A Write ACL is created using the bb3 role.

Module 9 – Delete ACL

A Delete ACL is created using the bb4 role.

7. System Design

User → Role → ACL → Table → Record

The system follows this access flow:

User



Assigned Role



ACL Evaluation



Script / Conditions



Record Access



Allow / Deny

The Read ACL is configured as a record-level ACL for the u_institution_details table, with the operation set to read.

8. Implementation

Step 1 – Create User

Navigate to:

User Administration → Users → New

Create the EEE User with the required details.

The PDF specifies:

User ID: EEE User

First Name: EEE

Last Name: User

Email: eeeuser@gmail.com

The screenshot shows the ServiceNow User Administration interface for a user named 'EEE User'. The page is divided into two main sections: user details on the left and system settings on the right. The user details section includes fields for User ID (EEE User), First name (EEE), Last name (User), Title, and Department. It also has checkboxes for 'Password needs reset', 'Locked out', and 'Active' (which is checked), and a dropdown for 'Identity type' set to 'Human'. The system settings section includes fields for Email (eeeuser@gmail.com), Language (None), Calendar integration (Outlook), Time zone (System (America/Los Angeles)), Date format (System (2026-08-19)), Business phone, and Mobile phone. At the bottom of the form are buttons for 'Update', 'Set Password', and 'Delete'. Below the form is a 'Related Links' section. The browser's address bar shows the URL 'dev442429.service-now.com/now/nav/ui/classic/params/target/sys_user.do%3Fsys_id%3D1c47356d47e38750c9e22afe016d4377%26sysparm_record_target%3Dsys_user%26...'. The browser's taskbar at the bottom shows the date and time as 11:47 on 26-09-2026.

User ID: EEE User

First name: EEE

Last name: User

Title:

Department:

Password needs reset: ☐

Locked out: ☐

Active: ☒

Identity type: Human

Internal Integration User: ☐

Email: eeeuser@gmail.com

Language: -- None --

Calendar integration: Outlook

Time zone: System (America/Los Angeles)

Date format: System (2026-08-19)

Business phone:

Mobile phone:

Photo: Click to add...

Update Set Password Delete

Related Links

Step 2 – Create Roles

Navigate to:

User Administration → Roles → New

Create:

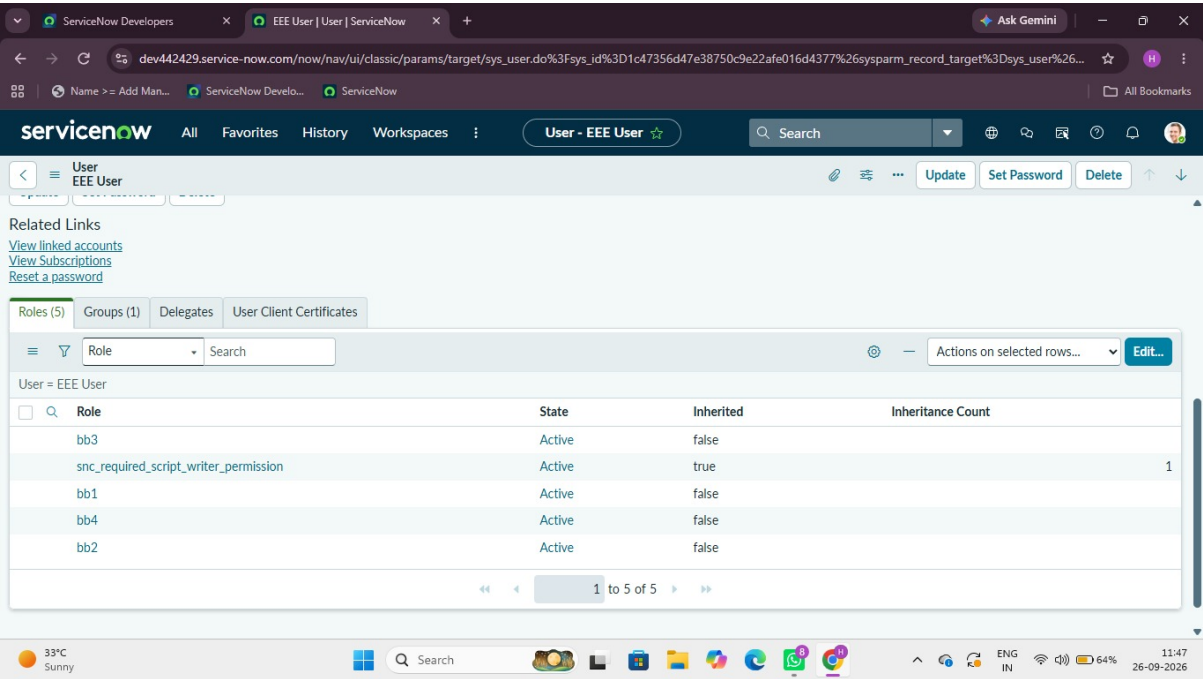
bb1

bb2

bb3

bb4

Assign the required roles to the EEE User.



Step 3 – Create Institution Details Table

Navigate to:

System Definition → Tables → New

Enter:

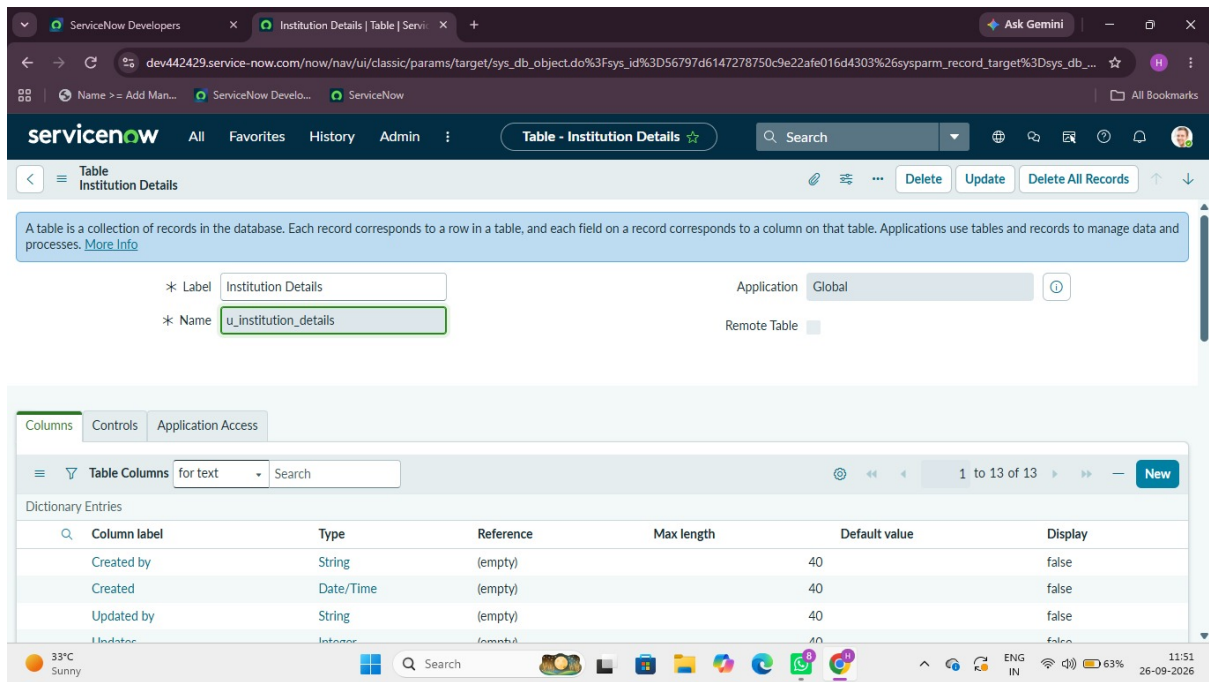
Label: Institution Details

Name: u_institution_details

Extends table: False

ACL script

Table Creation



Step 4 – Create Fields

Add the required fields to the table:

Field Type

Student Roll Number Auto Number

Student Name Reference – User

Faculty Name Reference – User

Branch Choice

Email String

Phone Number String

Description Multi String

For Branch, add:

ECE, EEE, CSE

Step 5 – Create Records

Create multiple student records with different Branch values such as:

ECE

EEE

CSE

This is required to test whether the ACL correctly controls record access.

9. ACL Script

Navigate to:

System Security → Access Control (ACL)

Elevate the Security_admin role and create a new Read ACL.

Configuration:

Type: Record

Operation: Read

Name: u_institution_details

Active: True

Advanced: True

Required Role: bb1

The PDF also specifies a data condition related to the Branch field and provides the following script.

```
(function () {  
  
    // Allow admin users full access  
    if (gs.hasRole('admin')) {  
        return true;  
    }  
  
    // Allow only users with bb1 role  
    if (gs.hasRole('bb1')) {  
        return true;  
    }  
  
    // Deny access for all others  
    return false;  
  
})();
```

The script checks the user's role. If the user has the administrator role, access is allowed. If the user has the bb1 role, access is also allowed. Otherwise, access is denied.

Read ACL Configuration

The screenshot shows the ServiceNow Access Controls configuration interface. The browser address bar displays a URL with a long alphanumeric string. The ServiceNow header includes navigation links like 'All', 'Favorites', 'History', and 'Workspaces', along with a search bar and a 'ServiceNow' logo. The main configuration area is titled 'Access Control u_institution_details'. It contains several fields: 'Type' set to 'record', 'Operation' set to 'read', 'Application' set to 'Global', 'Active' checked, and 'Advanced' unchecked. The 'Decision Type' is set to 'Allow If'. The 'Protection policy' is set to '-- None --'. The 'Name' field is set to 'Institution Details [u_institution_details]'. The 'Description' field contains the text: 'Allow read for records in u_institution_details, for users with role u_institution_details_user.' Below the main configuration area, there is a 'Conditions' section with a 'Requires role' field. The bottom of the screen shows a Windows taskbar with various application icons, a search bar, and system status indicators like temperature (33°C), time (12:29), and date (26-09-2026).

10. Create, Write and Delete ACL

Create ACL

Configuration:

Type: Record

Operation: Create

Name: u_institution_details

Active: True

Required Role: bb2

No Data Condition is required according to the given PDF.

Write ACL

Configuration:

Type: Record

Operation: Write

Name: u_institution_details

Active: True

Required Role: bb3

Delete ACL

Configuration:

Type: Record

Operation: Delete

Name: u_institution_details

Active: True

Required Role: bb4

Create ACL

ServiceNow Developers

u_institution_details | Access Co

Ask Gemini

dev442429.service-now.com/now/nav/ui/classic/params/target/sys_security_ad.do%3Fsys_id%3Da0c9b5a147278750c9e22afe016d4343%26sysparm_record_target%3Dsys_se...

Name >= Add Man... ServiceNow Develo... ServiceNow

All Favorites History

Access Control - u_institution_details

Search

Update Delete

* Type

record

* Operation

create

Admin overrides

☒

Decision Type

Allow If

Protection policy

-- None --

* Name

Institution Details [u_institution_details]

Description

Allow create for records in u_institution_details, for users with role u_institution_details_user.

Application

Global

Active

☒

Advanced

☐

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.

2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

33°C Sunny

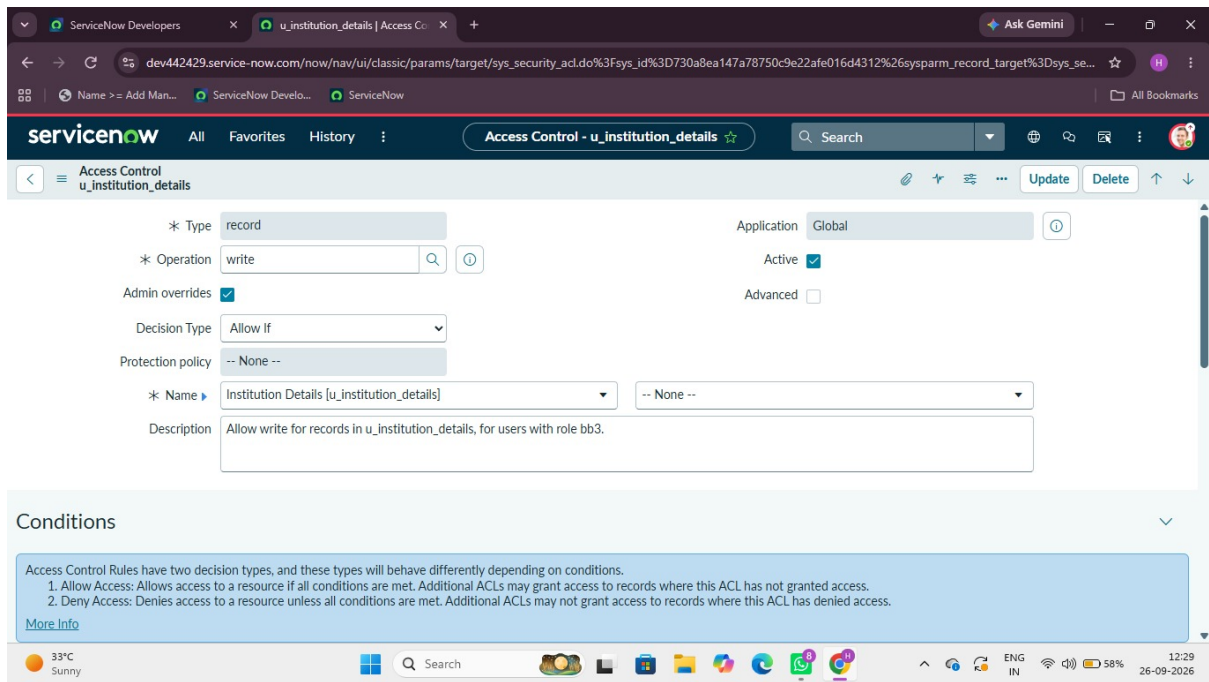
Search

ENG IN

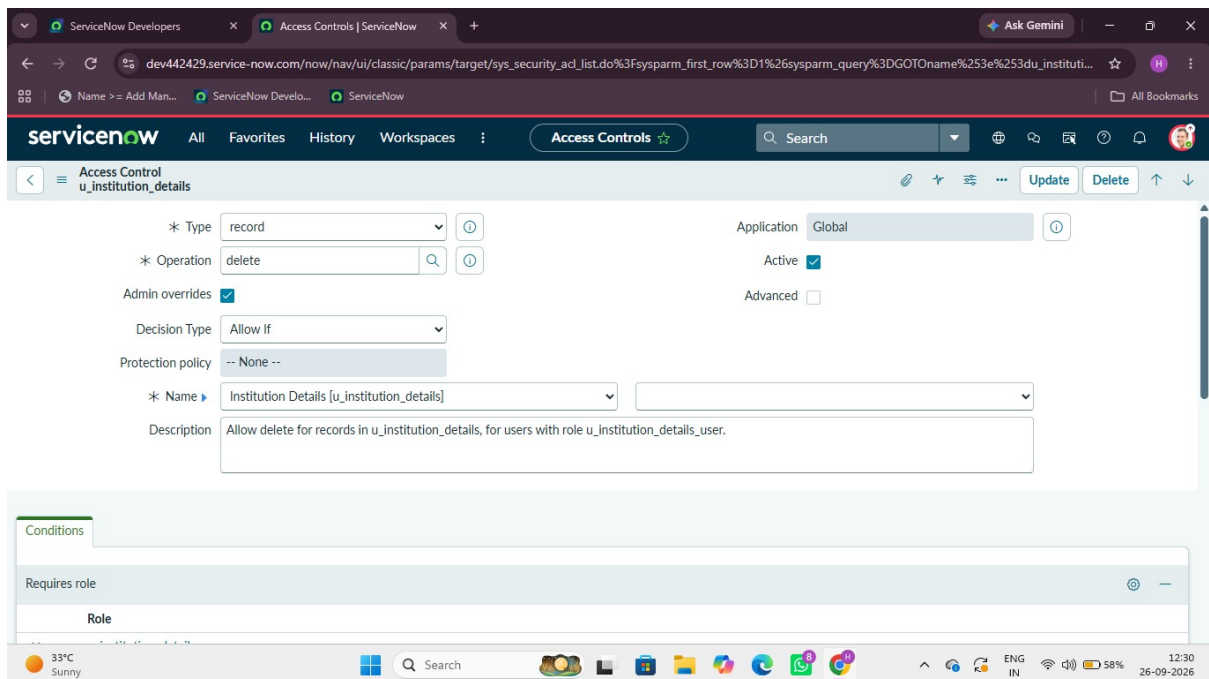
58%

12:30 26-09-2026

Write ACL



Delete ACL



11. Screenshot Demonstration

The system is tested by impersonating different users and opening the Student Records list.

For the Read ACL, the PDF instructs testing with the EEE user and Admin user and verifying the access behavior.

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Allyson Gillispie		Alejandra Prenatt	STU200

The nine screenshots included in this documentation demonstrate:

1. User creation

2. Role creation

3. Table creation

4. Field creation

5. Student records

6. Read ACL and script

7. Create ACL

8. Write ACL

9. Delete ACL

12. Testing

Test Case	User/Role	Operation	Expected Result
TC01	Admin	Read	All records accessible
TC02	bb1	Read	Read access according to ACL
TC03	No required role	Read	Access denied
TC04	bb1 + bb2	create	Create option available
TC05	bb1 + bb2 + bb3	write	Edit access available
TC06	bb1 + bb2 + bb3 + bb4	delete	Delete access available

The PDF confirms that users with bb1 can view the relevant records, users without the role cannot view them, and administrators can view all records.

It also states that adding bb2, bb3, and bb4 progressively enables Create, Write, and Delete operations.

13. Advantages

- Provides record-level security.
- Prevents unauthorized access.
- Supports role-based access control.
- Allows different permissions for different operations.
- Provides administrator-level access.
- Protects records from unauthorized modification and deletion.
- Improves data security within ServiceNow.
- ACLs can control access across forms and lists.

14. Limitations

- ACL configuration requires proper ServiceNow permissions.
- Incorrect ACL configuration may prevent legitimate users from accessing records.
- Multiple ACLs can make security configuration difficult to manage.
- Users must be assigned the correct roles.
- Testing is required after changing ACL rules.

15. Applications

- This type of ACL-based security can be applied in:
- Student Management Systems
- Employee Management Systems
- Institution Management
- HR Applications
- Department-wise record management
- Enterprise ServiceNow applications

- Sensitive record management

16. Results

The project successfully demonstrates record-level access control using ServiceNow ACLs.

The implemented system provides different access levels using the roles bb1, bb2, bb3, and bb4. The Read ACL controls record visibility, while Create, Write, and Delete ACLs control the corresponding operations.

The final outcome demonstrates how ServiceNow can use user roles, field values, ACL conditions, and scripts together to protect records from unauthorized access. The PDF describes the overall outcome as enforcement of READ, WRITE, CREATE, and DELETE security at record level.

17. Conclusion

The Script-Controlled ACL – Restrict Record Access Based on Field Value project demonstrates how ServiceNow provides security through Access Control Lists.

A custom Institution Details table was created and populated with student records belonging to different branches. Custom roles were created and assigned to users. ACLs were then configured to control Read, Create, Write, and Delete operations.

The project helps in understanding how ServiceNow evaluates users, roles, field conditions, and scripts before providing access to records. Thus, ACLs provide an

effective mechanism for protecting sensitive information and preventing unauthorized operations.

18. Future Enhancement

- The system can be enhanced by:
- Adding more branches and departments.
- Creating more specific role-based permissions.
- Adding additional field-level security.
- Implementing more complex scripted conditions.
- Extending the ACL security to Service Portal applications.
- Adding automated testing for ACL conditions.
- Providing different access levels for different departments.